



Network Layer: Logical Addressing

IPv4 ADDRESSES

An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

- They are unique in the sense that each address defines one, and only one, connection to the Internet.
- The IPv4 addresses are universal in the sense that the addressing system must be accepted by any host that wants to be connected to the Internet.

IPv4 Address

- The IPv4 addresses are unique and universal.
- An IPv4 address is 32 bits long.
 - The address space of IPv4 is 2^{32} (4,294,967,296)
 - Notation.
 - Binary notation
 - Dotted-decimal notation

10000000 00001011 00000011 00011111

128.11.3.31

Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

Example 19.3

Find the error, if any, in the following IPv4

- a. 111.56.045.78*
- b. 221.34.7.8.20*
- c. 75.45.301.14*
- d. 11100010.23.14.67*

Solution

- a. There must be no leading zero (045).*
- b. There can be no more than four numbers.*
- c. Each number needs to be less than or equal to 255.*
- d. A mixture of binary notation and dotted-decimal notation is not allowed.*

Classful Addressing

- In classful addressing, the address space is divided into five classes: A, B, C, D, and E.
- Each of the IP address belongs to a particular class that's why they are classful

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Example 19.4

Find the class of each address.

- a. 00000001 00001011 00001011 11101111
- b. 11000001 10000011 00011011 11111111
- c. 14.23.120.8
- d. 252.5.15.111

Solution

- a. *The first bit is 0. This is a class A address.*
- b. *The first 2 bits are 1; the third bit is 0. This is a class C address.*
- c. *The first byte is 14; the class is A.*
- d. *The first byte is 252; the class is E.*

Classes and Blocks

- The classful addressing wastes a large part of the address space.
 - Class A:
 - Class B:
 - Class C:
 - Class D:

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

A block in class A address is too large for almost any organization. This means most of the addresses in class A were wasted and were not used.

- A block in class B is also very large, probably too large for many of the organizations that received a class B block.
- A block in class C is probably too small for many organizations.
- Class D addresses were designed for multicasting.
- The class E addresses were reserved for future use.

Structure of IPv4 Address

- Consists of Net ID and Host ID.

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

- Mask(Default Mask)
 - 32-bit number of contiguous 1's followed by contiguous 0's.
 - To help to find the net ID and the host ID.

Use of IPv4 Address

- **Subnetting**

- Divide a large address block into several smaller contiguous groups known as subnets.
- Use of flexible net mask as it increase no of 1 in mask.

- **Supernetting**

- Several networks are combined to create a supernetwork or supernet.
- To combine several contiguous address spaces into a larger single address space.
- For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks as one supernetwork.

Classless Addressing(CIDR)

- To overcome address depletion and give more organizations access to the Internet, classless addressing was designed and implemented.
- In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block of addresses.
- The size of the block varies based on the nature and size of the entity.
- For example, a household may be given only two addresses; a large organization may be given thousands of addresses.
- An ISP, as the Internet service provider, may be given thousands or hundreds of thousands based on the number of customers it may serve.

Restriction

- The addresses in a block must be contiguous.
- The number of addresses in a block must be a power of 2.
- The first address must be evenly divisible by the number of address.

Mask

- Consists of n consecutive 1's followed by zeros.
- n can be any number b/w 0 and 32.

Tips:

- In IPv4 addressing, a block of addresses can be defined as **x.y.z.t /n**, in which x.y.z.t defines one of the addresses and the /n defines the mask.
- The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.
- The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.
- The number of addresses in the block can be found by using the formula

$$2^{32-n}.$$

Classless Addressing: In this, Variable length blocks are used that belongs to no class.

Restrictions:

→ (i) Addresses in a Block must be Contiguous.

(ii) No. of Addresses in a Block must be a power of 2 [1, 2, 4, 8, 16, ...]

(iii) The first address must be evenly divisible by the no. of addresses.

Block → 205.16.37.32 → 16 = 2⁴ =
First → 205.16.37.33
 → 205.16.37.34

...
 Last → 205.16.37.47

Mask in classless Addressing can take any value from 0 to 32. (n) = 1, (32-n) = 0.

→ Slash/CIDR Notation.

x.y.z.t/n → defines the mask.

IMP Points:

(i) First Address in block can be found by setting the rightmost (32-n) bits to 0s.

(ii) Last address can be found by setting the rightmost (32-n) bits to 1s.

(iii) No. of Addresses: 2³²⁻ⁿ

Example: One of the Addresses is 205.16.37.39/28.

Find first, last and total no. of addresses.

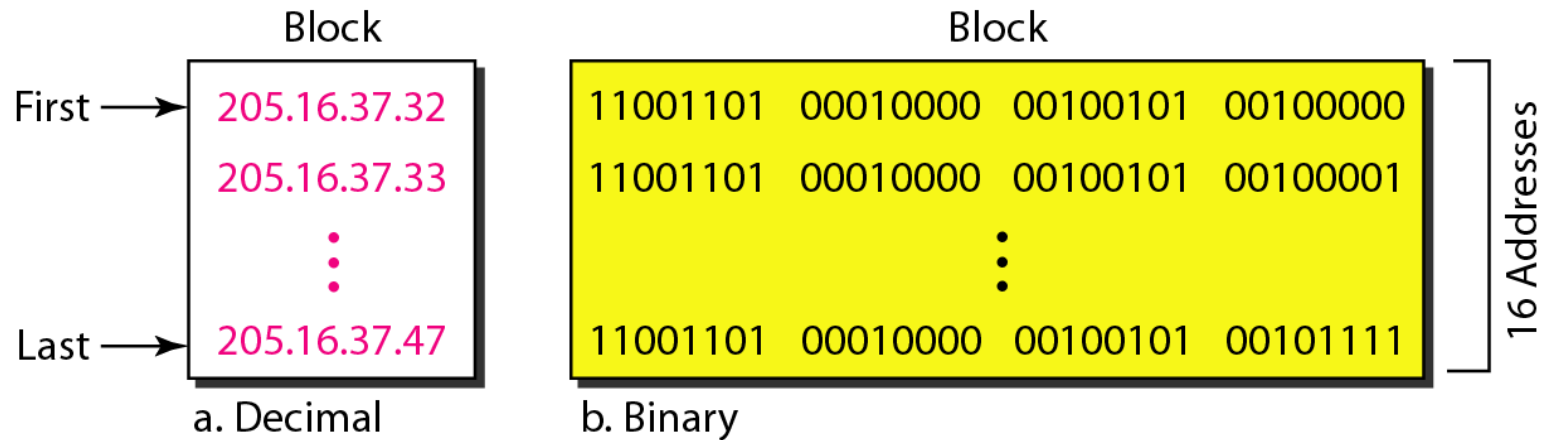
11001101 00010000 00100101 00100111 → 00101111
 F.A: ↓ ↓ ↓ 00100000 N.A: 2⁴
 = 205.16.37.32 L.A = 205.16.37.47 = 16

Example 19.5

Figure 19.3 shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.

We can see that the restrictions are applied to this block. The addresses are contiguous. The number of addresses is a power of 2 ($16 = 2^4$), and the first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Figure 19.3 *A block of 16 addresses granted to a small organization*



Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.

Example 19.7

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

This is actually the block shown in Figure 19.3.

Example 19.8

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.

Example 19.9

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Example 19.5 the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a. The first address**
- b. The last address**
- c. The number of addresses.**

Example 19.9 (continued)

Solution

- a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Example 19.9 (continued)

- b. The last address can be found by ORing the given addresses with the complement of the mask. Oring here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

Example 19.9 (continued)

- c. The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Mask complement: 00000000 00000000 00000000 00001111

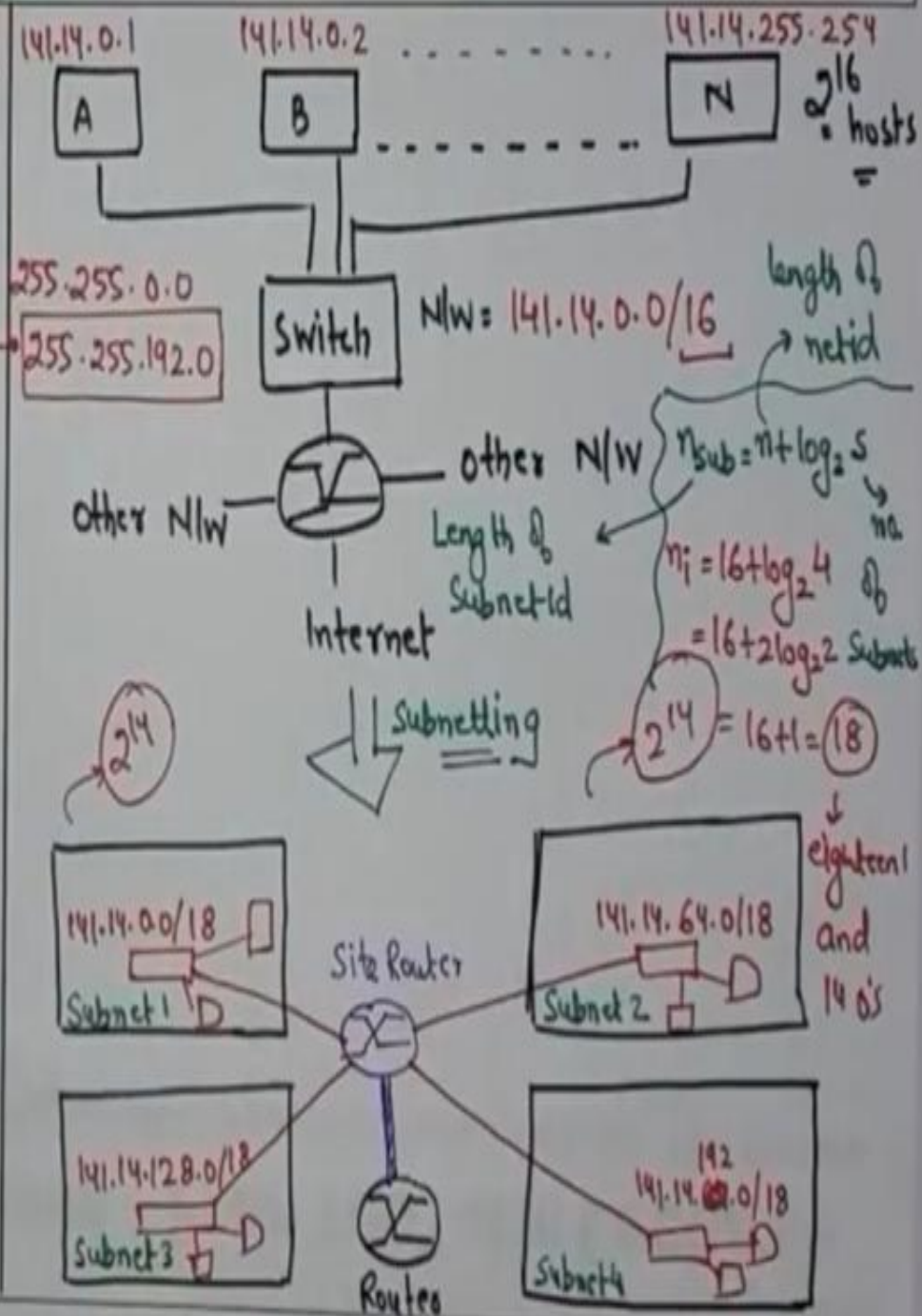
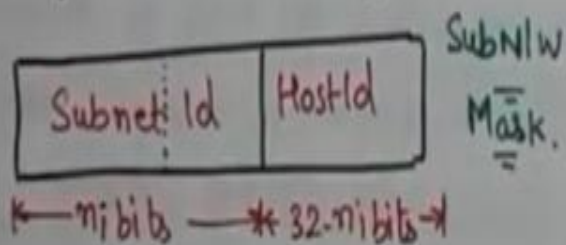
Number of addresses: $15 + 1 = 16$

Subnetting: - It is the idea of splitting a block to smaller blocks, In this a NW is divided into several smaller subnets (SubNWs) with each Subnetwork having its own Subnetwork address.

↳ A SubNW has Subnetid and Hostid.

[Subnetting $\uparrow$ - the length of netid and $\downarrow$ length of hostid or $\uparrow$ no. of 1s in the mask.]

Subnet Address: - When a NW is Subnetted, the first address in the Subnet is identifier of the Subnet and is used by the router to route the packets destined for that SubNW.

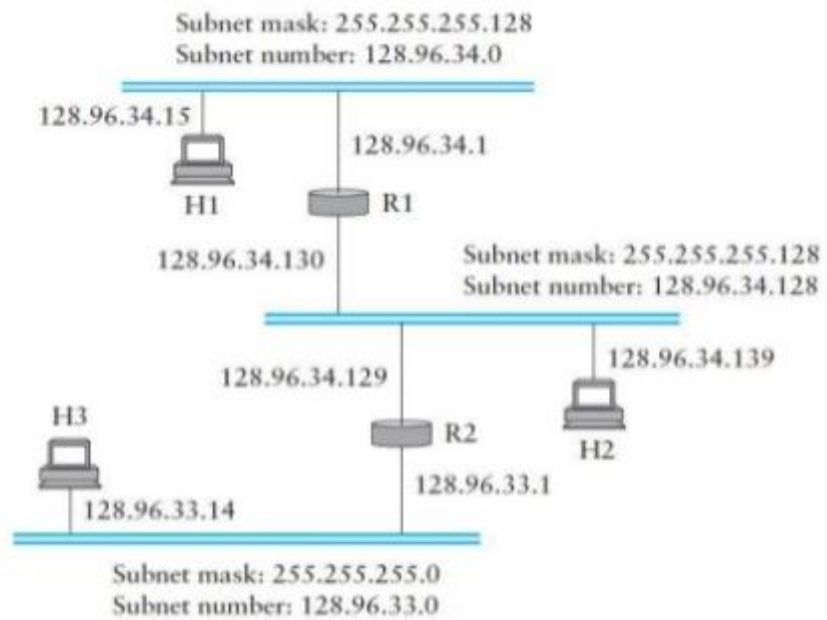


- In subnet the hosts may be on different physical networks but share a single network number.
- Each host in the subnet is configured with IP address and Subnet mask.
The subnet number is calculated by doing "bitwise AND" between IP address and subnet mask.

Example,

Create 6 subnets from the IP network address 192.168.1.128. Find the subnet mask and maximum no. of node per subnet and list of networks.

- $2^n = \text{No. of subnets}$. But we can use only $2^n - 2$ subnets only
- From given data,
- $2^n = 8$, so $n=3$ (3 bits needed from host id). But we can use 6 subnets.
- So total possible subnets are 8, from that 6 subnets can be used.



To find subnet mask,

Default subnet mask for class c address = 255.255.255.0

In binary = 11111111.11111111.11111111.00000000

We need 3 bits,

11111111. 11111111. 11111111. 11100000 => 255.255.255.224

So subnet mask is 255.255.255.224

To find Network ID,

Binary AND between IP address and subnet mask

192.168.1.128 => 11000000. 10101000. 00000001. 10000000

255.255.255.224 => 11111111. 11111111. 11111111. 11100000

192.168.1.128 => 11000000. 10101000. 00000001. 10000000

To find No. of hosts per subnet,

Balance bits are 5,

So $2^5 - 2 = 30$ **hosts** can be in single subnet.

Sub Network Number	Hosts		Broadcast Address	Remarks
	From	To		
192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31	Unusable
192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63	Usable
192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95	Usable
192.168.1.96	192.168.1.97	192.168.1.126	192.168.1.127	Usable
192.168.1.128	192.168.1.129	192.168.1.158	192.168.1.159	Usable
192.168.1.160	192.168.1.161	192.168.1.190	192.168.1.191	Usable
192.168.1.192	192.168.1.193	192.168.1.222	192.168.1.223	Usable
192.168.1.224	192.168.1.225	192.168.1.254	192.168.1.255	Unusable





